MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR TRANSISTOR
ARRAYS, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/108(USAF), dated 9 April 1976, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, line 2: Delete "Three" and substitute "Two".

1.2, delete and substitute:

"1.2 Part number. The part number shall be in accordance with MIL-M-38510."

PAGE 4

* TABLE I, delete the temperature coefficient of base emitter voltage test and substitute with revised test below:

*

Test	Symbol	Conditions	Device type	Limits 1/		Units
				Min	Max	
Temperature coefficient of base emitter voltage 4/ 8/	$\frac{\Delta V_{BE}}{\Delta T}$	V_{CE} = 3 V $-55^{\circ}C \leq T_A \leq +25^{\circ}C$	01	-2.2	-1.5	mV/°C
			02	-2.2	-1.3	
		I_E = -1 mA $+25^{\circ}C \leq T_A \leq +125^{\circ}C$	01	-2.2	-1.5	mV/°C
			02	-2.2	-1.3	

TABLE I, static forward current transfer ratio (beta), h_{FE} at I_C = 1mA, $+25^{\circ}C \leq T_A \leq +125^{\circ}C$; maximum limit column (device 02 only): Delete "300".

PAGE 5

3.6; delete and substitute:

"3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III."

TABLE II, delete and substitute:

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"TABLE II. Electrical test requirements

MIL-STD-883 test requirement	Class S Devices	Class B Devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5005)	1*, (2, 3, 4)**	1*, (2, 3, 4)**
Group A test requirements (method 5005)	1, 2, 3, 4, 5, 6, 7, 9, 10, 11	1, 2, 3, 4
Group C end point and group B Class S electrical parameters (method 5005)	1, 2, 3 and Table IV delta limits	Table IV delta limits and limits
Additional electrical subgroups for group C periodic inspection	None	5, 6, 7, 9, 10, 11

* PDA applies to subgroup 1 (see 4.3e)

** $\frac{\Delta V_{BE}}{\Delta T}$ and $\frac{\Delta \lvert V_{BEQA}-V_{BEQB} \rvert}{\Delta T}$ tests as specified in table III herein for group A, subgroups 2 and 3, and f_t tests as specified in table III herein for group A, subgroup 4, are not required for final electrical tests (for device 02 only) but shall be performed for group A sample testing."

PAGE 6

*3.7, delete second sentence in its entirety.

PAGE 24

TABLE III, tests 68 through 72, maximum limits column: Delete "300".

PAGE 26

*TABLE III, tests 119 through 123, maximum limits column: Delete "-1.5" and substitute "-1.3".

PAGE 26

TABLE III, tests 128 through 132, maximum limits column: Delete "300".

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*TABLE III, tests 156 through 160, maximum limits column: Delete "-1.5" and substitute "-1.3".

PAGE 30

*4.5.2, line 2 following cooled to: Add "within 10°C of their power stable condition,".

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TABLE IV, delete and substitute:

* TABLE IV. Groups B and C end point electrical parameters (T_A = 25°C)

TEST	Device types 01 and 02	
	Limit	Delta
V_{BE}	.600 V min .800 V max	±.010V
$\lvert V_{BEQA} - V_{BEQB} \rvert$	2.0 mV	
h_{FE}	70 min 300 max 1/	±10%

1/ The 300 max limit applies to device 01 only.*

4.6, delete.

6.4, line 2: Delete "MIL-STD-1313, MIL-STD-1331, MIL-S-19500E," and substitute "MIL-M-38510, MIL-STD-1331, MIL-S-19500,".

6.6; delete and substitute:

"6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed Generic-Industry type. Generic-Industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of Generic-Industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-Industry type
01	3018A
02	3045

Note: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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